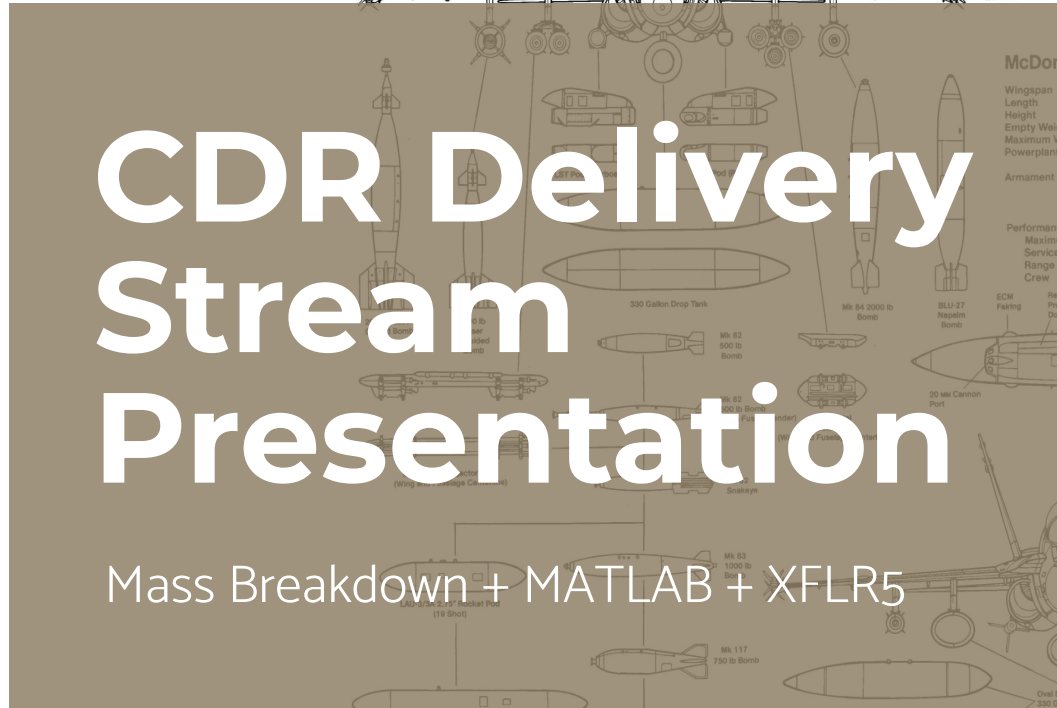
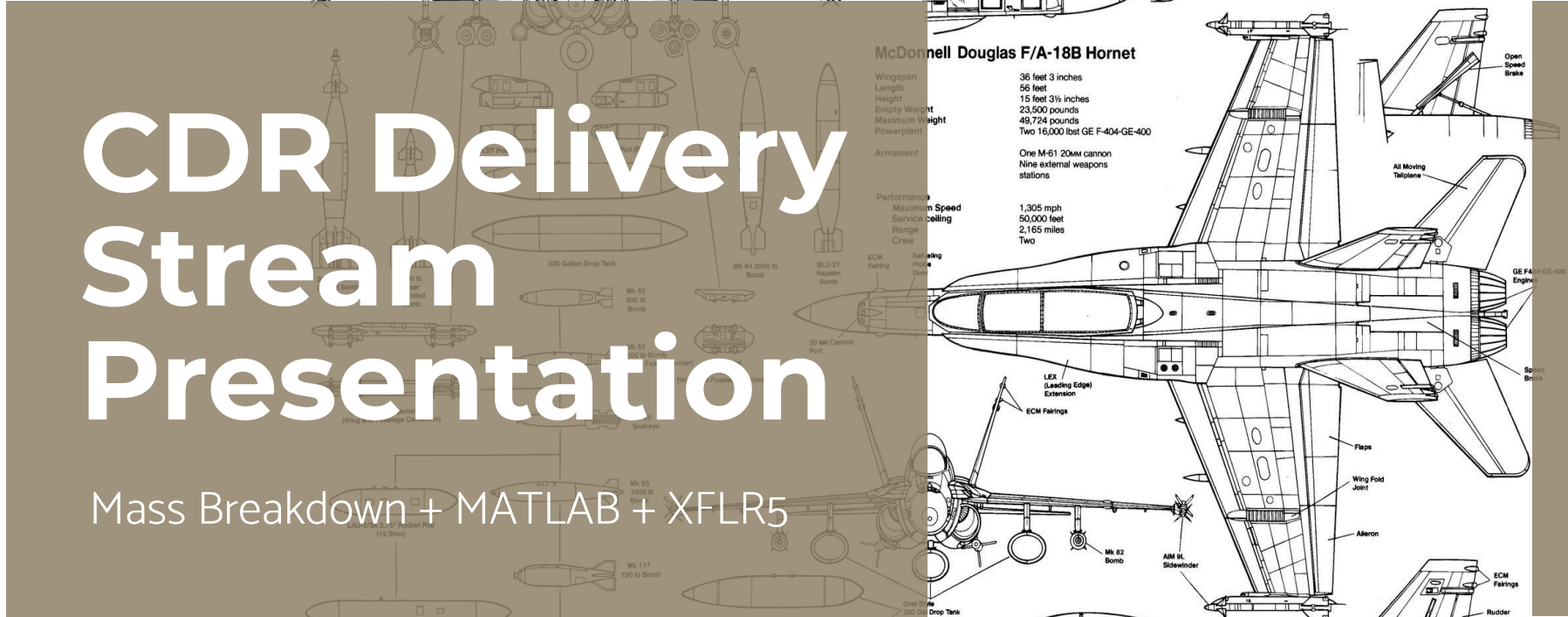




CDR Delivery Stream Presentation

Mass Breakdown + MATLAB + XFLR5

Mass Breakdown + MATLAB + XFLR5





Weight Of The Design

02

Revised Mass Breakdown Using CAD-Derived BOMs

Component	Mass	Notes
Main wing	251.80g	
Wing holder and payload holder	182.28g	
Fuselage	486.30g	
Horizontal stabilizer	99.00g	
Vertical stabilizer	77.14g	
Tail holder	131.19g	
Landing gear + wheels	Front landing gear = 320.66g Rear landing gear = 56.41g	
Adhesives + coverings + fasteners + miscellaneous	100g	Estimated

Revised Mass Breakdown Using CAD-Derived BOMs

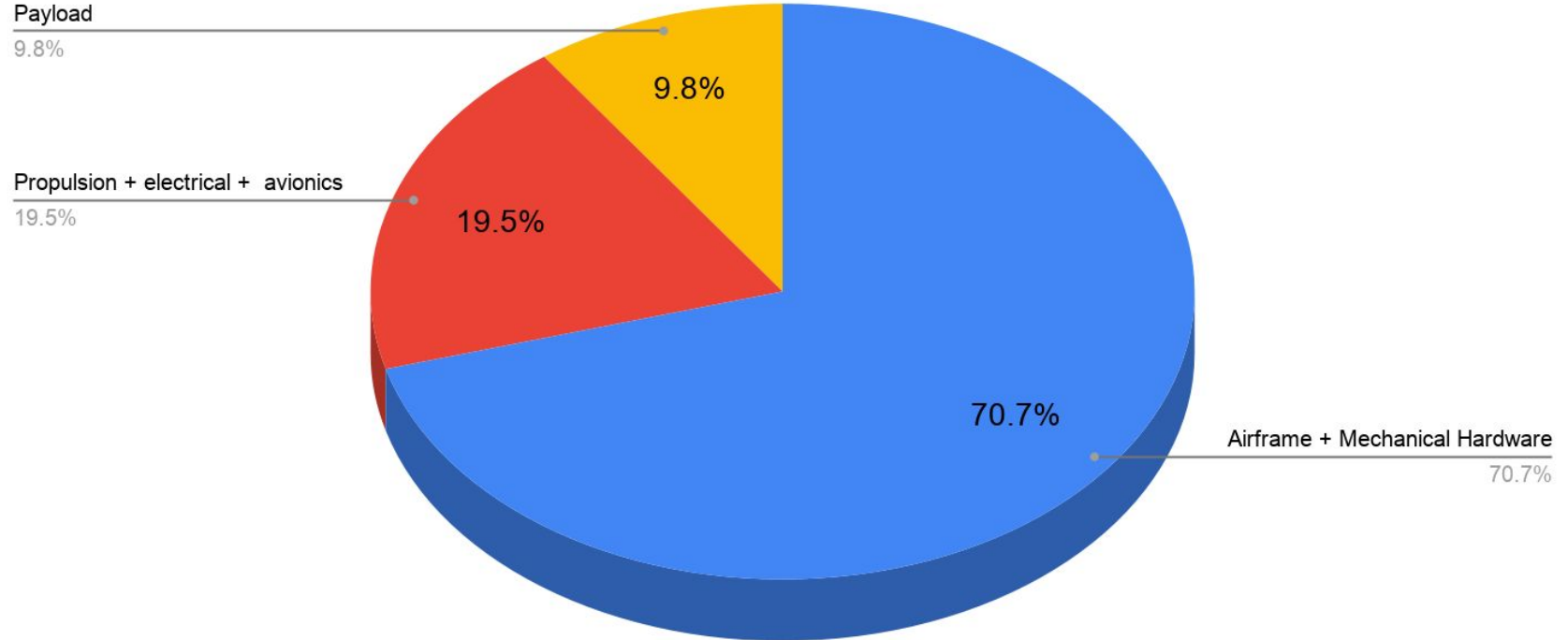
Component	Mass	Notes
Motor	113g	SunnySky X2814-class motor data lists ~110 g for a similar 2814 motor
Propeller	22g	APC 12 x 4.5 propeller listed at 0.78 oz $\approx$ 22 g.
ESC	45g	Estimated
Battery	170g	Compared to the listed 3S 2200 mAh LiPo listed at 165.56g
Servos	36g	9g per servo
Wiring + connectors	35g	Estimated
Guidance system	50g	Estimated
Motor	113g	SunnySky X2814-class motor data lists ~110 g for a similar 2814 motor
Payload	237g	8 oz of water

Total Mass

Category	Mass
MTOW	2.41kg
Minimum operating weight (no payload)	2.17kg

Category	Mass	% of MTOW
Airframe structure + mechanical hardware	1.705kg	70.66%
Propulsion + electrical + avionics	0.471kg	19.54%
Payload	0.237kg	9.83%

CDR Mass Breakdown Pie Chart

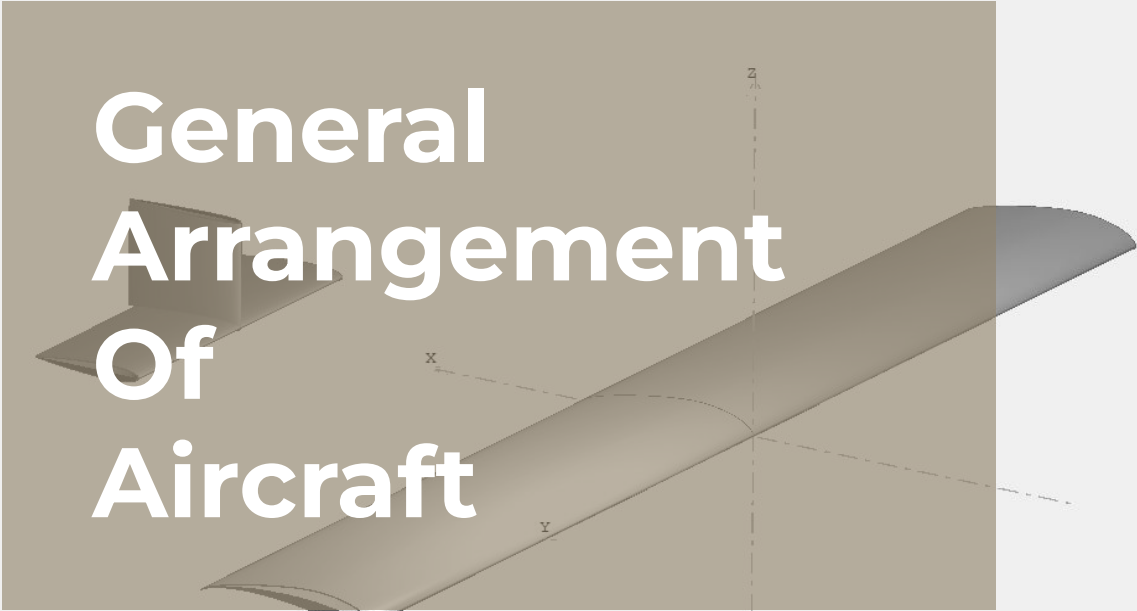


Component	X Position (m)	Y Position (m)	Z Position (m)
Propeller	-0.400	0.000	-0.060
Motor	-0.380	0.000	-0.060
Wiring + connectors	-0.225	0.000	-0.060
ESC	-0.200	0.000	-0.060
Battery	-0.200	0.000	-0.060
Landing gear (front wheels)	-0.150	0.000	-0.150
Adhesives + fasteners (around wing CG)	0.000	0.000	-0.060
Right wing aileron servo	0.000	0.750	0.000
Left wing aileron servo	0.000	-0.750	0.000
Fuselage (approx. midpoint)	0.025	0.000	-0.060
Payload	0.060	0.000	-0.060
Self-guidance allowance	0.300	0.000	-0.060
Back wheel for landing gear	0.400	0.000	-0.100
Wing/Payload holder	0.650	0.000	0.000
Elevator servo	0.720	0.000	0.000
Rudder servo	0.720	0.000	0.000
Tail holder	0.720	0.000	0.000

NOTE:

- 1.) Fuselage is 120cm in length
- 2.) Main wing LE is located 30 cm behind the nose
- 3.) Tail LE is 0.720m behind the main wing LE
- 4.) Payload mounted near quarter-chord point to minimize CG variation during unloaded configurations

General Arrangement Of Aircraft

A 3D CAD model of an aircraft's wing and tail section. The wing is a straight, rectangular planform with rounded leading and trailing edges. The tail consists of a vertical stabilizer and a horizontal stabilizer. A 3D Cartesian coordinate system is shown with the Z-axis pointing vertically upwards, the X-axis pointing horizontally to the left, and the Y-axis pointing diagonally downwards and to the right. The model is rendered in a semi-transparent brown color.

```
Cruise config
Wing Span      = 1.500 m
xyProj. Span   = 1.500 m
Wing Area      = 0.321 m²
xyProj. Area    = 0.321 m²
Plane Mass     = 1.900 kg
Wing Load      = 5.919 kg/m²
Tail Volume    = 0.472
Root Chord     = 0.214 m
MAC            = 0.214 m
TipTwist       = 0.000°
Aspect Ratio   = 7.009
Taper Ratio    = 1.000
Root-Tip Sweep = 0.000°
XNP = d(XCp.Cl)/dCl = 0.095 m
Mesh elements  = 1875
```

Airfoil Selection

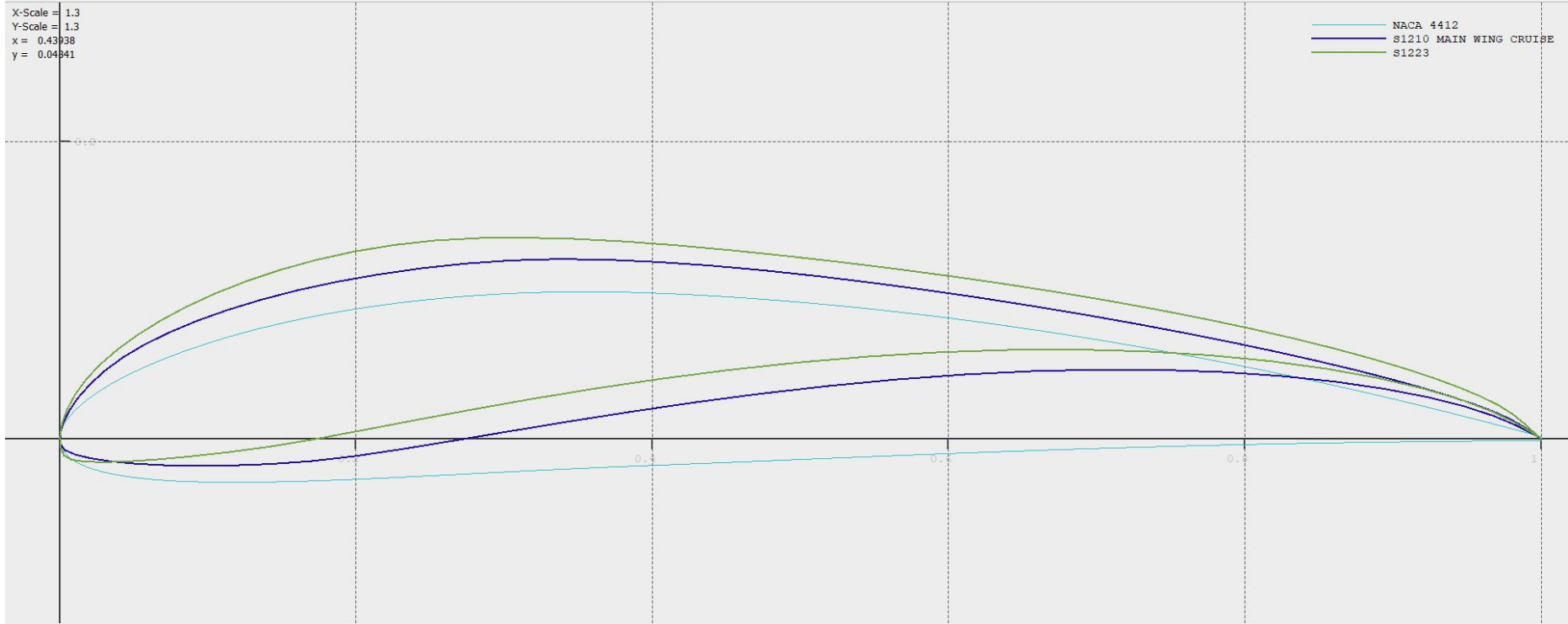
- The airfoil for the main wing should provide low-speed lift, efficiency, and manufacturability to support stability during the low-speed nature of the mission
- NACA and non-NACA airfoils were examined through XFLR5
- NACA 4412 failed to meet the Cl requirements of 1.5, and S1233's camber made manufacturability a concern, hence the variation S1210 was chosen

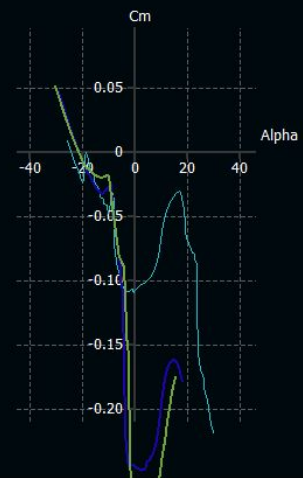
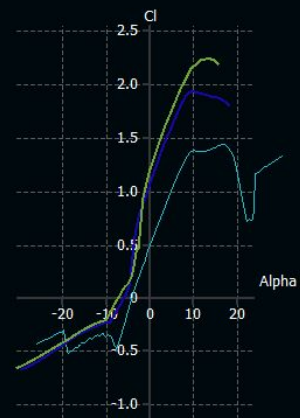
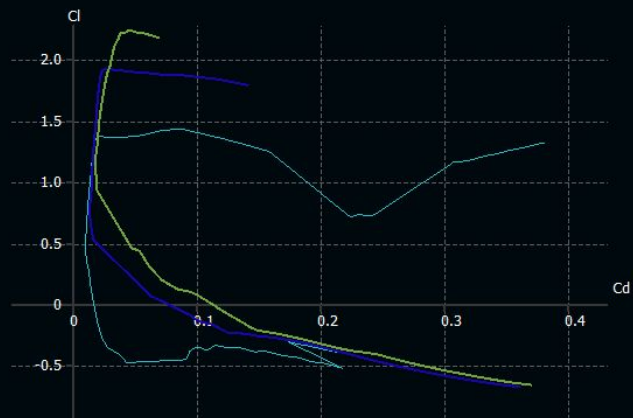
Metric	NACA 4412	S1233	S1210
Max Lift	Medium	High	High
Cruise efficiency	Medium	High	High
Manufacturability	High	Low	Medium
Final Decision	Rejected	Rejected	Accepted

- NACA 0012 were kept as the airfoil for horizontal and vertical stabilizers

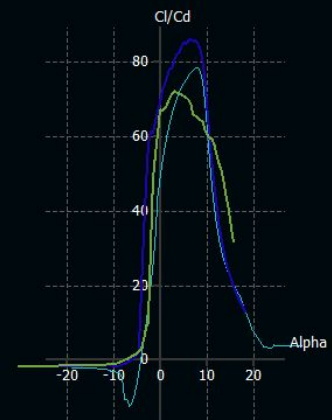
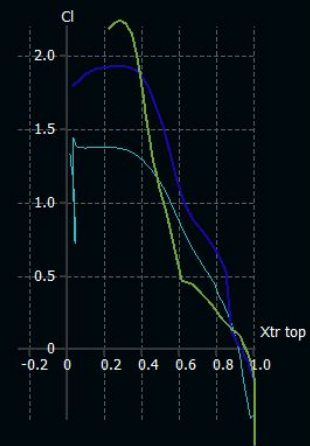
X-Scale = 1.3
Y-Scale = 1.3
x = 0.43038
y = 0.04841

NACA 4412
S1210 MAIN WING CRUISE
S1223



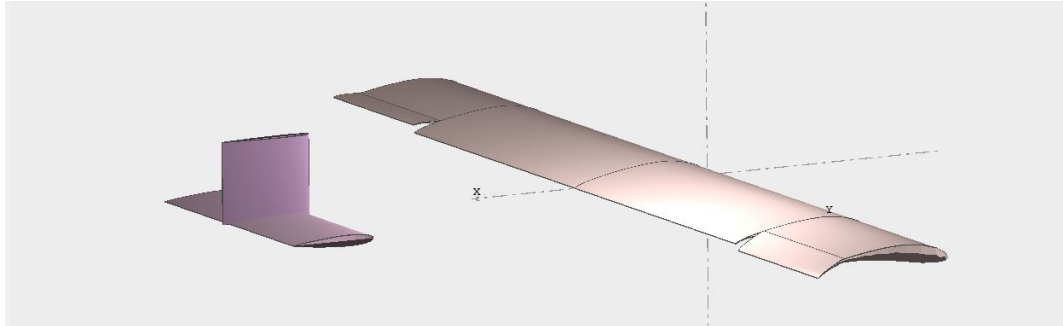


NACA 4412
 T1_Re0.200_M0.00_N9.0
 S1210 MAIN WING CRUISE
 T1_Re0.200_M0.00_N9.0
 S1223
 T1_Re0.200_M0.00_N9.0



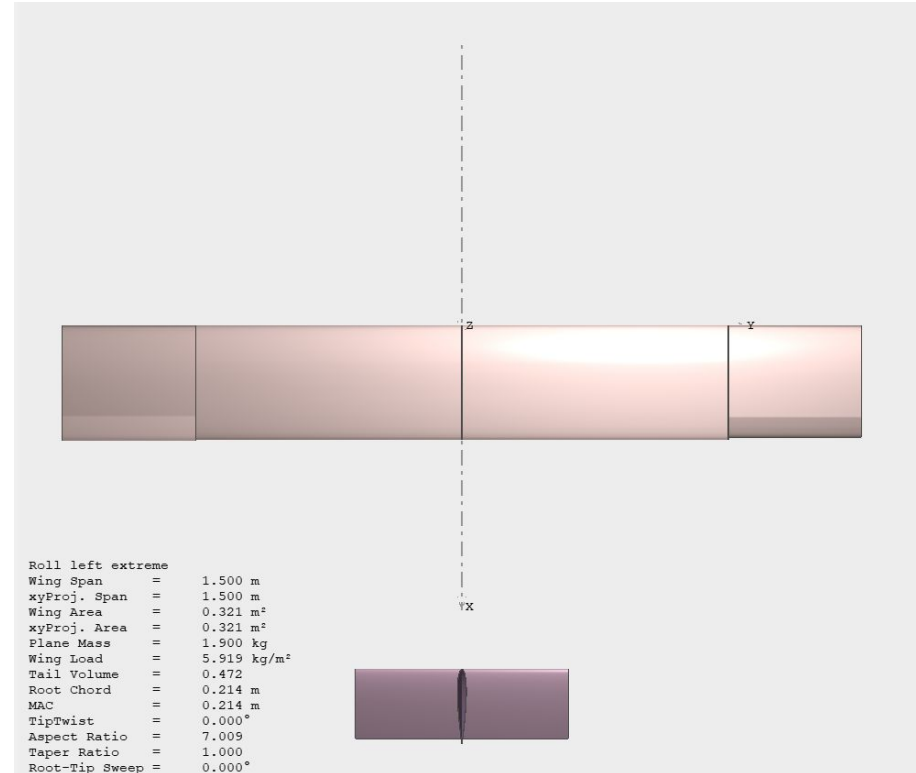
Main Wing Configuration

Row	y (m)	Chord (m)	Foil	Meaning
1	0.00	0.214	S1210 cruise config	start of wing
2	0.50	0.214	S1210 cruise config	end of wing
3	0.50	0.214	S1210 aileron	start of aileron
4	0.75	0.214	S1210 aileron	end of aileron



Main Wing Configuration Cont'd

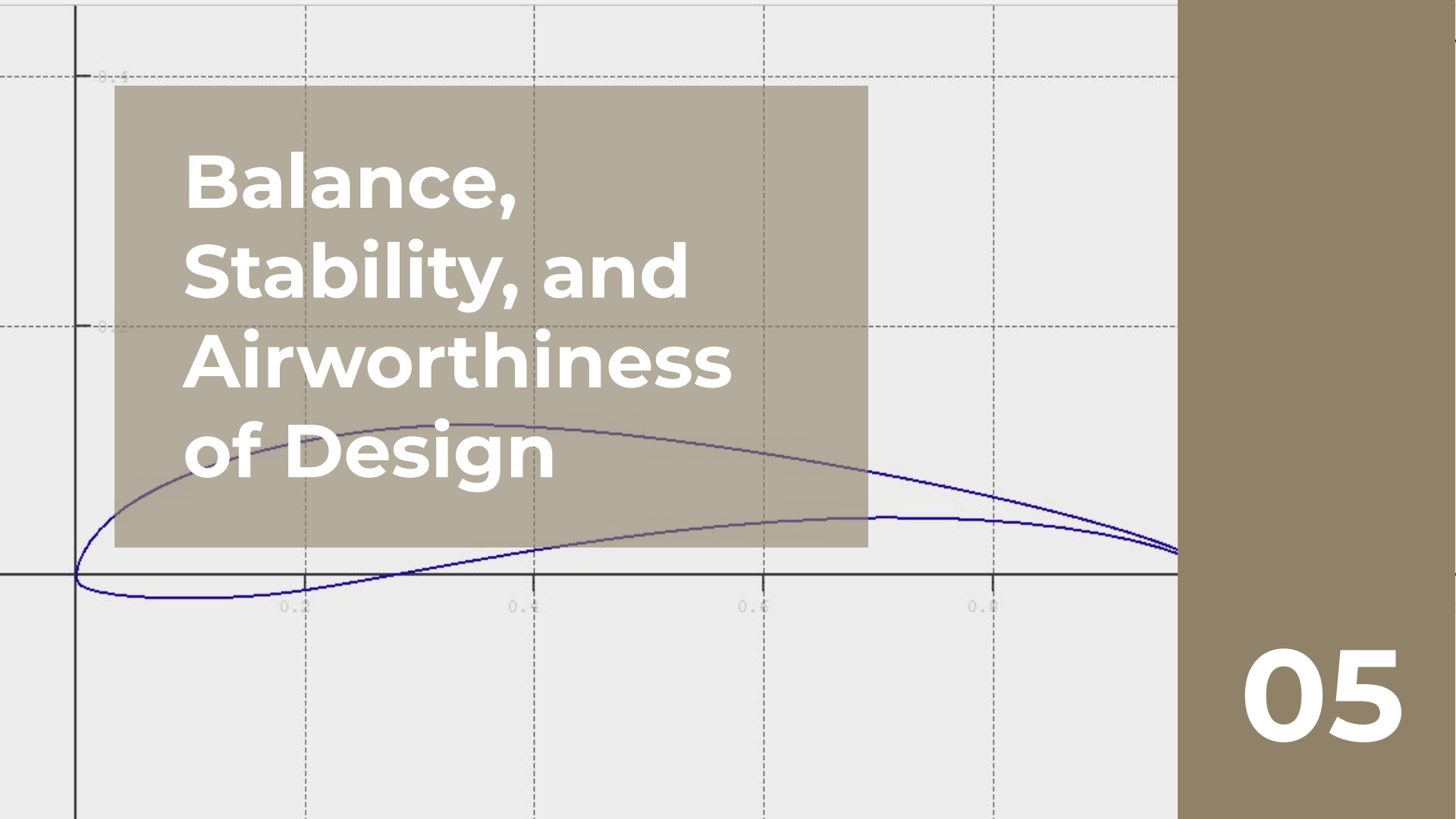
Main Wing Parameters	Values
Wingspan	1.5 m
Wing area	0.321 m ²
Aspect ratio	7
Wing loading	58.046 N/m ²
Root chord	0.214 m
Tip chord	0.214 m
Taper ratio	1
Offset	0
Dihedral	0°
Twist	0°



Control Surface Sizing

Parameters	Values
Ailerons	20% chord at 0.25m in span

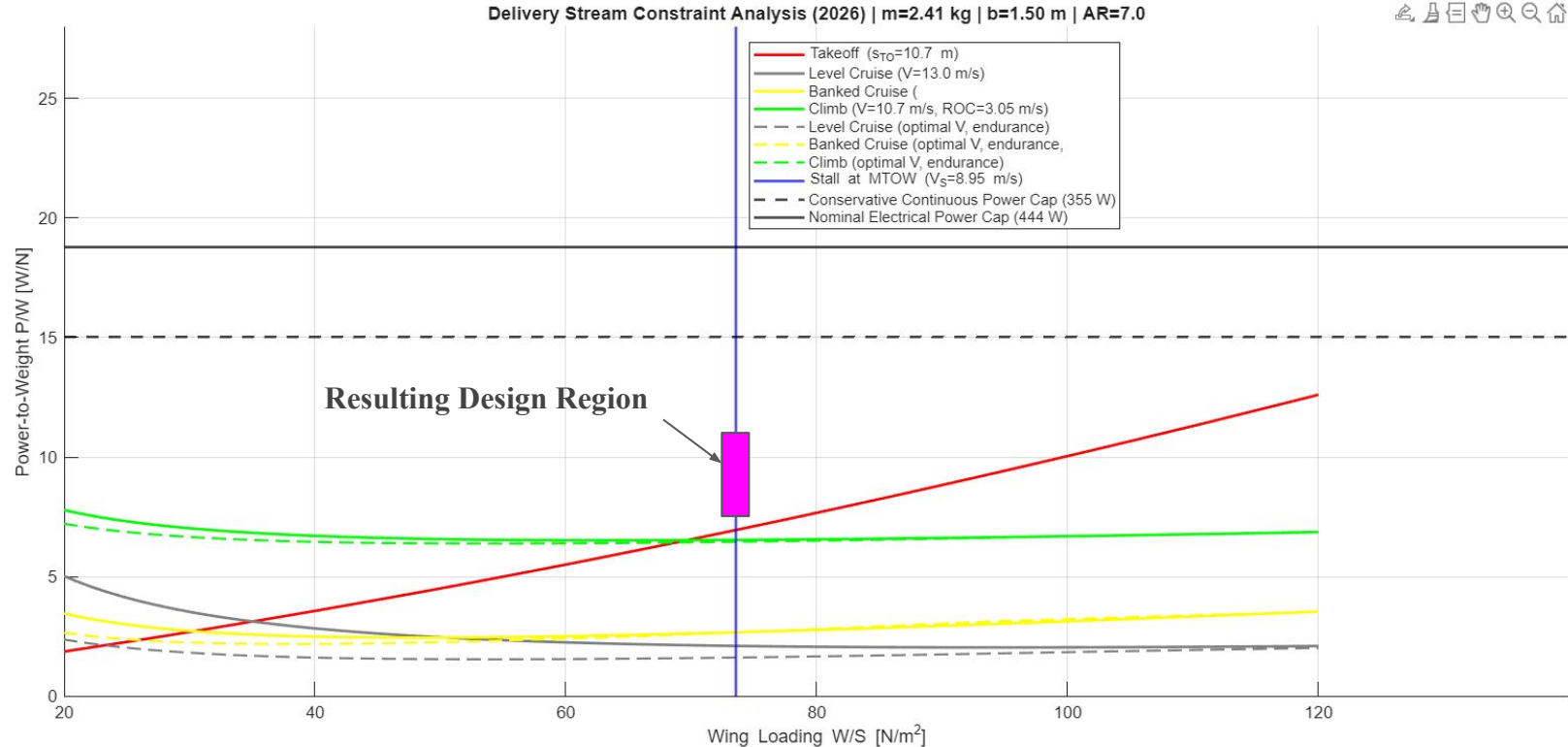
Surface	Span / height	Root chord	Tip chord	Control surface
Horizontal tail	0.50 m	0.130 m	0.130 m	<u>Elevator</u> : 25% chord
Vertical tail	0.24m	0.140 m	0.140 m	<u>Rudder</u> : 30% chord



Balance, Stability, and Airworthiness of Design

05

Revised MATLAB Analysis Using Revised Mass Breakdown



Revised MATLAB Analysis Using Revised Mass Breakdown

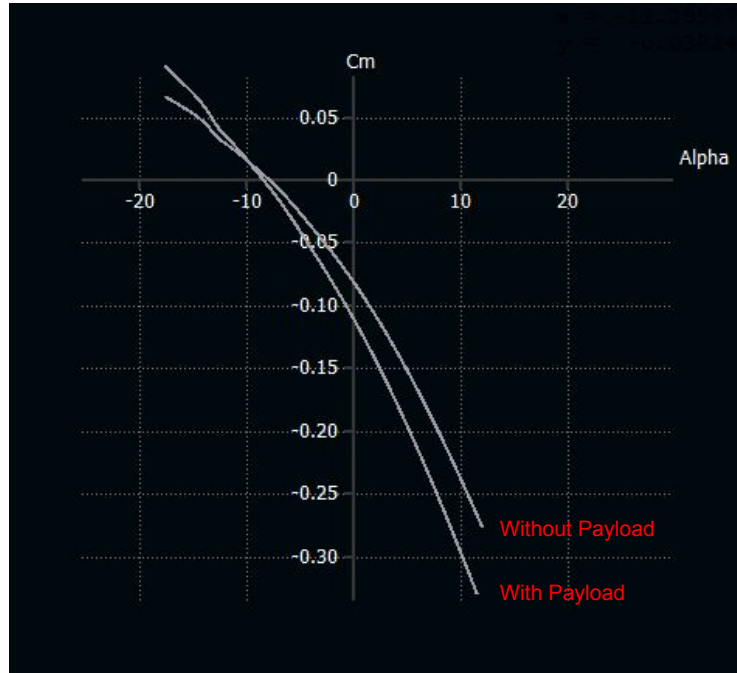
- MATLAB analysis uses 2.41kg as the new MTOW, which yields a **maximum wing loading of $W/S = 73.57\text{N/m}$ and a minimum P/W of 6.95W/N , which aligns with XFLR5 wing loading (sanity check)**
- This implies that the aircraft approaches the stall wing-loading limit at takeoff and turns
- **However, this does not mean the aircraft will stall**
- $P_{TO} = (P/W) \times \text{Weight} \rightarrow P = (6.95\text{W/N})(23.64\text{N}) \rightarrow P \approx 164\text{W}$
- $P_{\text{nominal}} = VI = (11.1\text{V})(40\text{A}) = 444\text{W} \rightarrow P_{\text{consistent}} = 0.80(444\text{W}) = 355\text{W}$
- **At a 6.95W/N , there is a 216% power margin relative to the largest consistent power output available (estimated as 80 percent of max power available)**
- Thus, the propulsion system should be more than adequate
- **The stall margin should still be considered, ensuring max speed during takeoff and banked turns**

Important Design Metrics Updated

Metric	Value
CDR Design Mass	2.41 kg
Takeoff speed	10.7 m/s
Wing span	1.50 m
Aspect ratio	7
Wing area	0.321 m ²
Mean Chord	0.214 m
Wing loading from XFLR5	73.6 N/m²
P/W Ratio from MATLAB	At minimum 6.5W/N and at most 15W/N

Metric	Value
Cruise Speed	13 m/s
Bank Speed	11 m/s
Stall Speed	8.95 m/s
Bank Angle	45°
Climb Speed	10.7 m/s
Required ROC	3.05 m/s
Clean C _{L max}	1.50
Takeoff C _{L max}	1.60
Conservative Power Cap	355 W -> 80% of max electric cap
Max Power Cap	444 W -> P=VI = (11.1 V)(40 A)

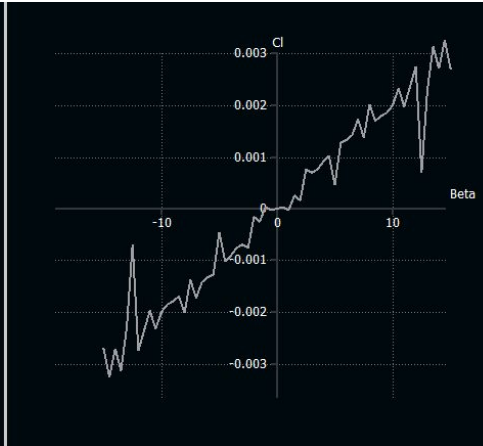
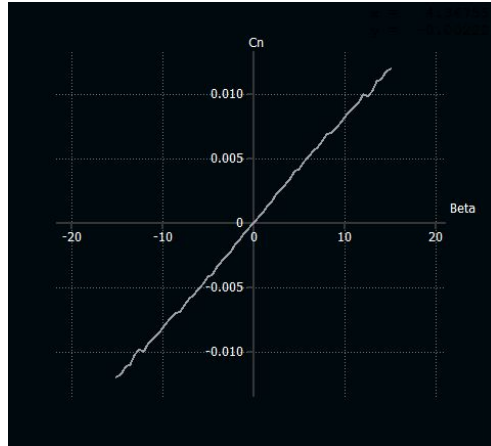
Aircraft Pitch Stability For All Configurations



Parameter	Value
Analysis Method	VLM2
Configuration	Cruise
Neutral Point	0.118 m
CG With Payload	0.078 m
CG without Payload	0.080m
MAC	0.214 m
Static Margin With Payload	17.80%
Static Margin No Payload	16.80%

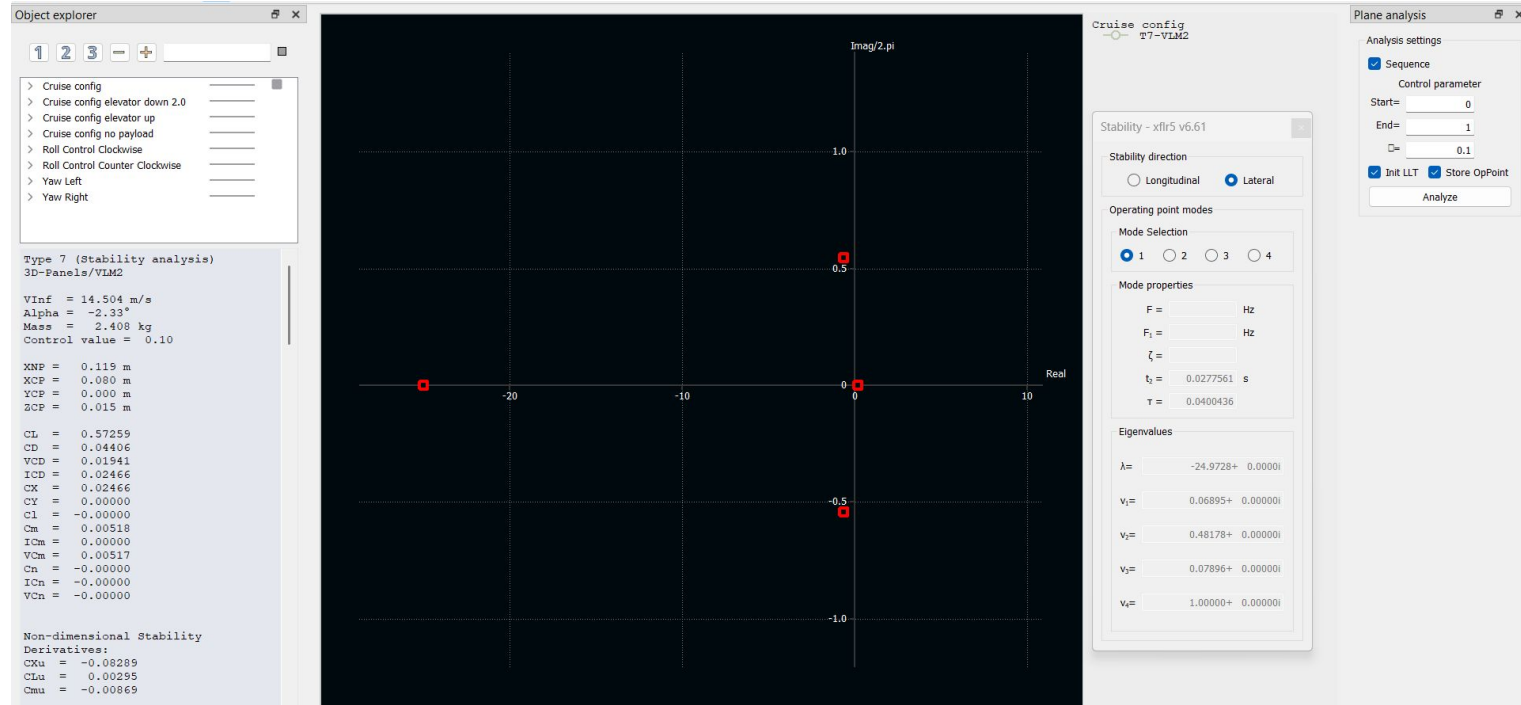
Aircraft Directional Stability

Parameter	Cn Values	Cl Values
$\beta = 15^\circ$	$C_n = +0.01195$	$Cl = +0.00270$
$\beta = 0^\circ$	$C_n = 0.000$	$Cl = 0.000$
$\beta = -15^\circ$	$C_n = -0.01195$	$Cl = -0.00270$

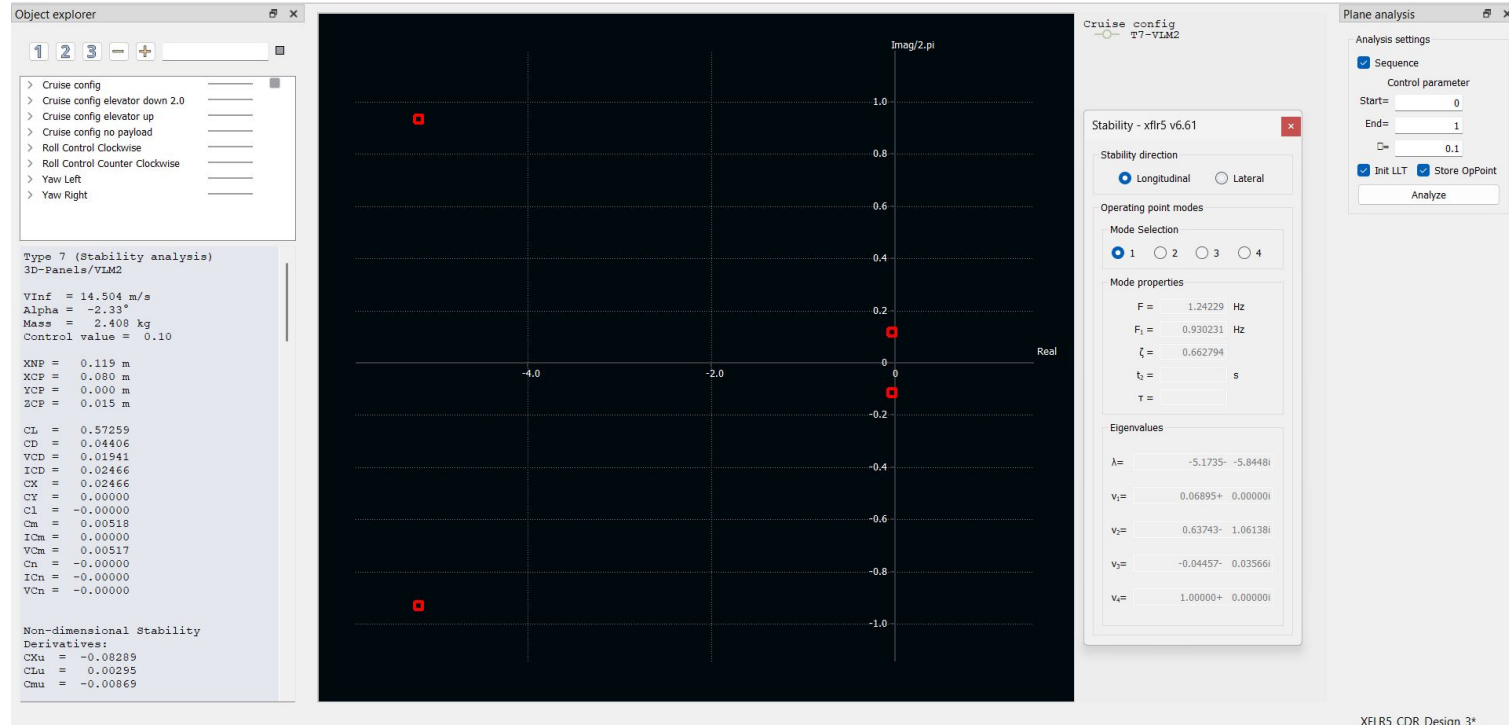


- C_n vs Beta plot is nominal with a strong upward trend
- Cl vs Beta shows a very shallow positive slope, which can be simplified as a horizontal line (slope = 0.00054)
- This implies nominal yaw stability and a neutral lateral stability, which is expected with 0 dihedral

Aircraft Dynamic Longitudinal Stability



Aircraft Dynamic Lateral Stability



Important Dynamic Stability Notes

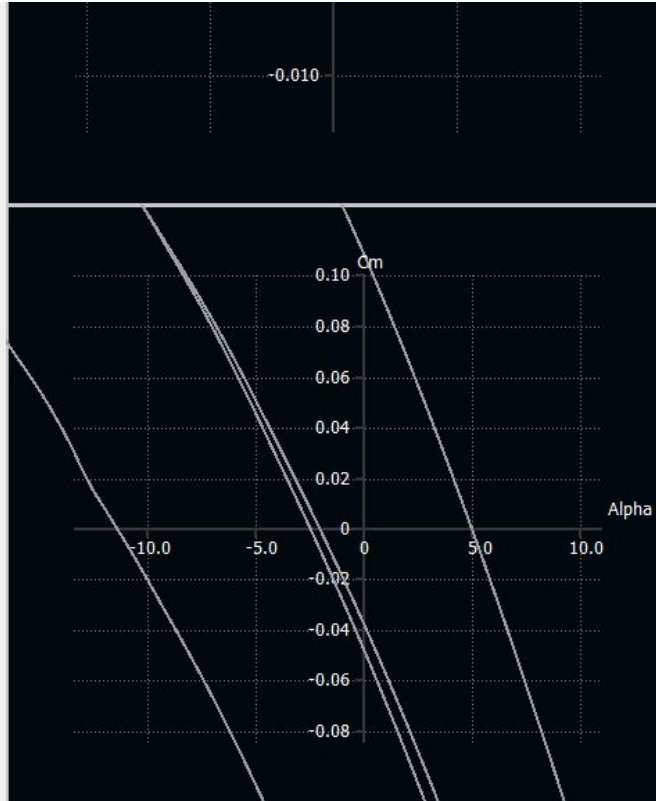
Parameter	Important Notes Regarding Results
Longitudinal	Longitudinal modes have negative real components, while modes 3 to 4 are lightly damped but stable.
Lateral	Lateral modes also have negative real components, while modes 2 and 3 have low but effective dampening. Mode 4 is slightly positive which indicates light spiral, but with a small and manageable magnitude with pilot control
Resulting Cruise Metrics for Dynamic Stability Analysis	- Cruise speed of 14.5m/s, which is acceptable and close to original CDR value - Cruise alpha of -2.3°, expected with cambered airfoils

Elevator Trim Requirements: $\pm 5^\circ$

```
Type 1: Fixed speed
VInf = 13 m/s
3D-Panels/VLM2
Using plane inertia
Mass = 2.408 kg
CoG.x = 0.07775 m
CoG.z = -0.0504 m

B.C. = Dirichlet
Analysis type = Viscous
Ref. dimensions = Projected
Ref. area = 0.321m²
Ref. span = 1.500m
Ref. chord = 0.214m
Density = 1.225 kg/m³
Viscosity = 1.5e-05m²/s

Data points = 60
```

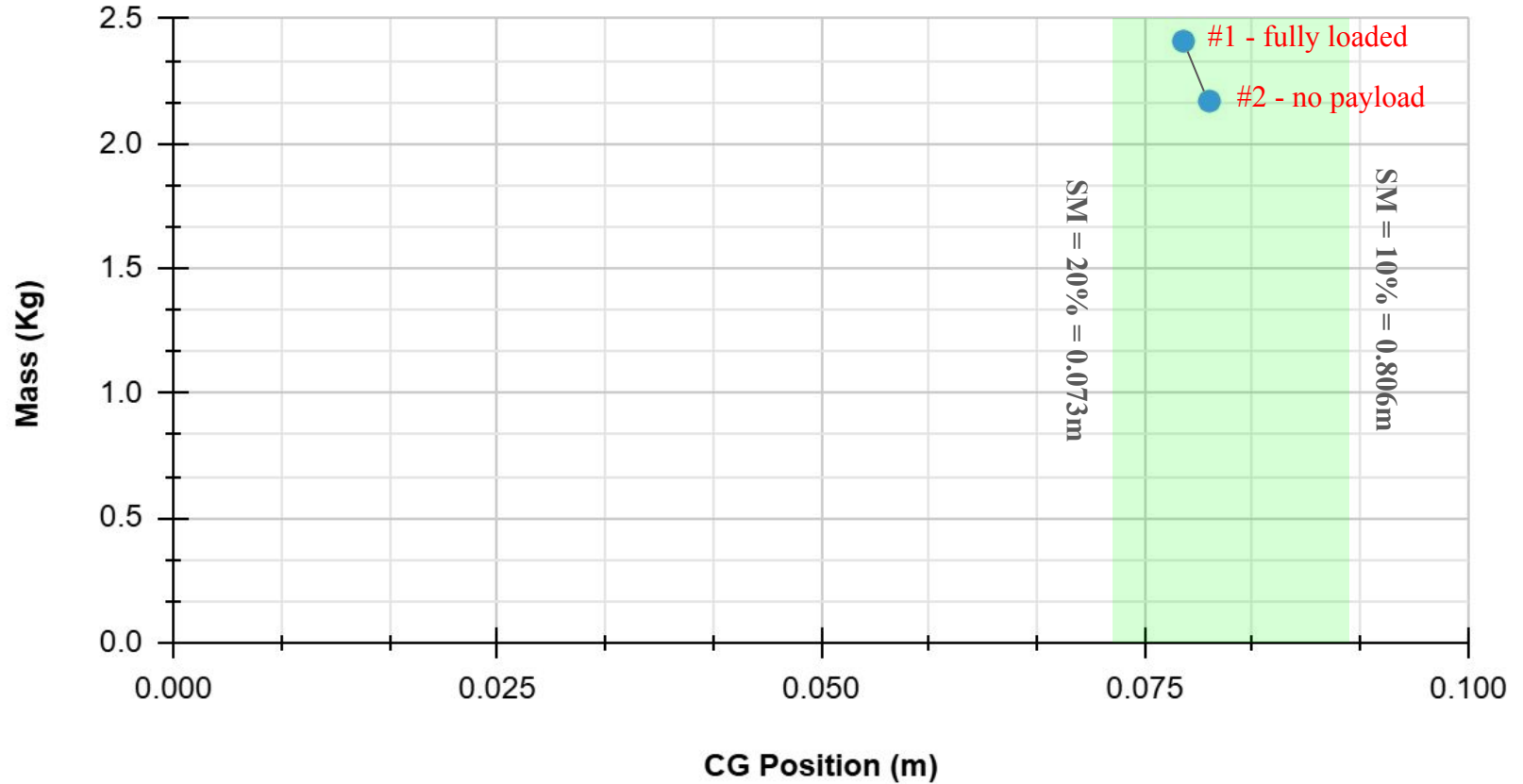


- The aircraft is trimmable, using an elevator deflection of $\pm 5^\circ$
- This is evident between an AOA interval of -12° to $+5^\circ$

CG Envelope Table

Parameter	CGX Value With Respect to the LE of the Main Wing	Corresponding SM
Forward limit	0.073 m	20%
Fully Loaded: MTOW	0.078 m	17.80%
Empty: No Payload	0.080 m	16.80%
Aft limit	0.095m	10%

CG Envelope



Note: Neutral Point is located at 0.118m

Delivery Time Estimate

$$t_{\text{delivery}} \approx t_{\text{TO}} + t_{\text{climb}} + t_{\text{approach}} + d_{\text{LZ}} / V_{\text{CRUISE}}$$

$$V_{\text{Stall}} = 8.23 \text{ m/s}$$

$$V_{\text{CRUISE}} = 13.0 \text{ m/s}$$

$$\text{ROC} = 3.05 \text{ m/s}$$

$$\text{Runway distance} = 10.7 \text{ m}$$

Takeoff:

$$V_{\text{LOF}} = 1.1V_{\text{Stall}} \text{ to } 1.2V_{\text{Stall}} = 9.05 \text{ to } 9.88 \text{ m/s}$$

$$t_{\text{TO}} = 2(\text{takeoff distance}) / V_{\text{LOF}} \approx 2.2 \text{ to } 2.4 \text{ s}$$

Climb:

Assume climb altitude = 6 to 9 m

$$t_{\text{climb}} = h / \text{ROC} \approx 2.0 \text{ to } 3.0 \text{ s}$$

Approach:

$$V_{\text{app}} = 1.2V_{\text{S}} \text{ to } 1.3V_{\text{S}} = 9.88 \text{ to } 10.70 \text{ m/s}$$

Assume final approach distance = 30 to 45 m

$$t_{\text{approach}} \approx 2.8 \text{ to } 4.6 \text{ s}$$

Total:

$$t_{\text{delivery}} \approx (7 \text{ to } 10 \text{ s}) + d_{\text{LZ}} / 13.0$$

where d_{LZ} is the unknown distance to the landing zone in metres.